

Test of Hypothesis
Part-2
(Small Samples Tests)
By
Dr. Padigepati Naveen

Recall... Hypothesis testing steps

Use of the following sequence of steps in applying hypothesis testing.

1. From the problem context, identify the parameter of interest and an appropriate test statistic .
2. State the null hypothesis, H_0 .
3. Specify an appropriate alternative hypothesis, H_1 .
4. Choose a significance level .
5. State the rejection region for the statistic.
6. Compute any necessary sample quantities, substitute these into the equation for the test statistic, and compute that value.
7. Decide whether H_0 should be rejected or not *and report that in the problem context.*

Reasons to small samples

- ❑ The value of a statistics obtained from a large sample is generally close to the parameter of the population.
- ❑ But there are situations when one has to take a small sample.
- ❑ E.g. if a new medicine is to be introduced, a doctor cannot test the new medicine by giving it to many patients.
- ❑ Thus he takes a small sample (i.e., sample size $n \leq 30$).
- ❑ When the sample is small, then sampling distribution of statistic will not follow normal distribution (i.e., *central limit not applicable in these cases*).
Then distributions like t , F etc. are used for testing.

Small Samples Test's

Under small sample tests, we will see important tests. Test the significance of

1. Single mean
2. Difference of means
3. Chi-square test for goodness of fit & independence of attributes.

Point to note... Degree of freedom:

- ❑ The Degree of freedom is the number of independent observations of the variable.
- ❑ But The number of independent observations is different for different statistics.
- ❑ Suppose we are asked to select any five observation. There is no restriction on the selection of these observations. Hence degree of freedom is 5.
- ❑ Suppose we want to select five observations whose sum is 100. Here four observations can be can be selected freely but the 5th observation is automatically selected by the restriction of total 100.

Point to note...

- ❑ We are not free to select all the five observations but our freedom is restricted to the selection of only 4 observations.
- ❑ Thus the degree of freedom for selecting n observation when one such restriction is given is n-1.
- ❑ If two such restrictions are given the degree of freedom will be n-2.
- ❑ The degree of freedom associated with some of the important statistics are given below:

$$(a) D.f \text{ of } \bar{x} = \frac{\sum x_i}{n} \text{ is } n$$

$$(b) D.f \text{ of } S^2 = \frac{\sum (x_i - \bar{x})^2}{n} \text{ is } n-1$$

Student's t-distribution

Student's t-distribution:

- This is very important distribution was given by W.S.Gosset in 1908.
- He published his work under the pen-name of student.
- Hence the distribution is known as student's t-distribution.

If $x_1, x_2, \dots, x_n$ is a random sample of 'n' observations drawn from a normal population with mean μ and S.D. σ then the student's "t" is defined by the statistic

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

and follows t-distribution with $v = n-1$ degree of freedom.

- Where $\bar{x} = \frac{\sum x_i}{n}$ and $s^2 = \frac{\sum (x_i - \bar{x})^2}{n-1}$

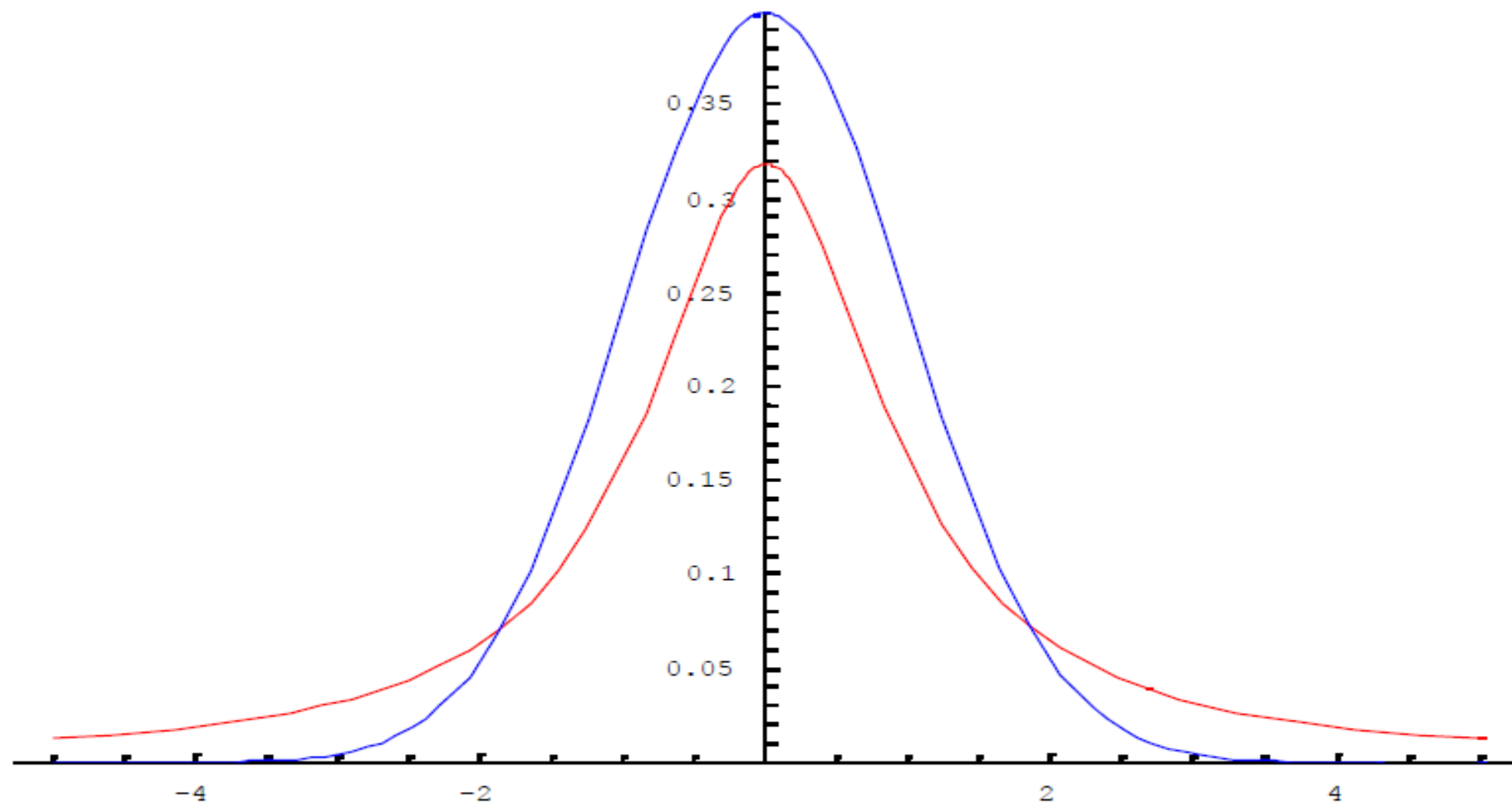
Assumption of t-distribution:

- 1) The population from which the sample is drawn is normal.
- 2) The sample is random and size $n \leq 30$.
- 3) The population S.D. σ is not known.

Properties of t-distribution:

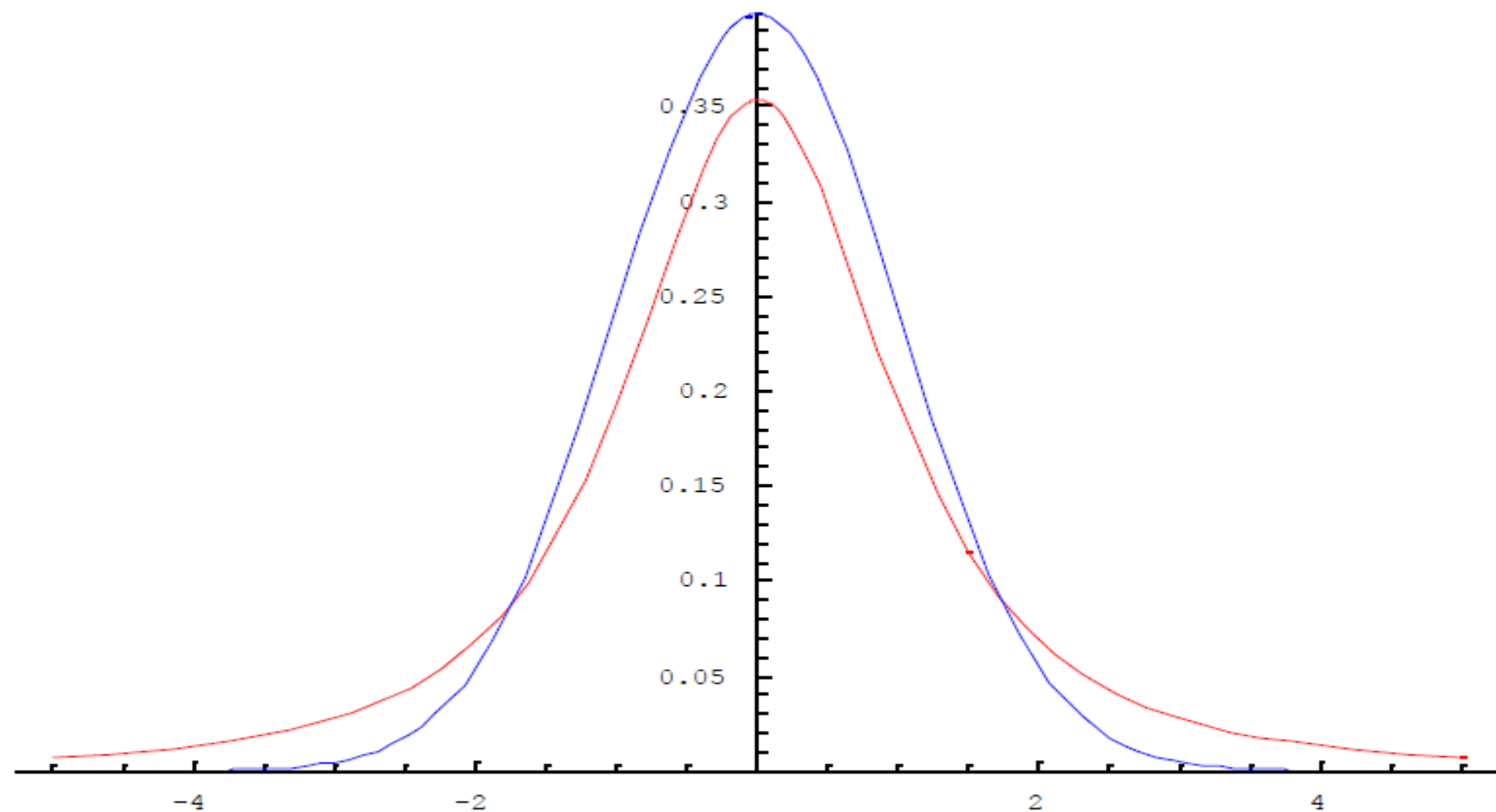
- 1) The probability curve of t-distn. is symmetrical.
- 2) The tails of the curve are asymptotic to x-axis.
- 3) When $n \rightarrow \infty$, t-distn tends to normal distribution.
- 4) The form of the t-dist. Varies with the degrees of freedom.

Student T Distribution with 1 degrees of freedom



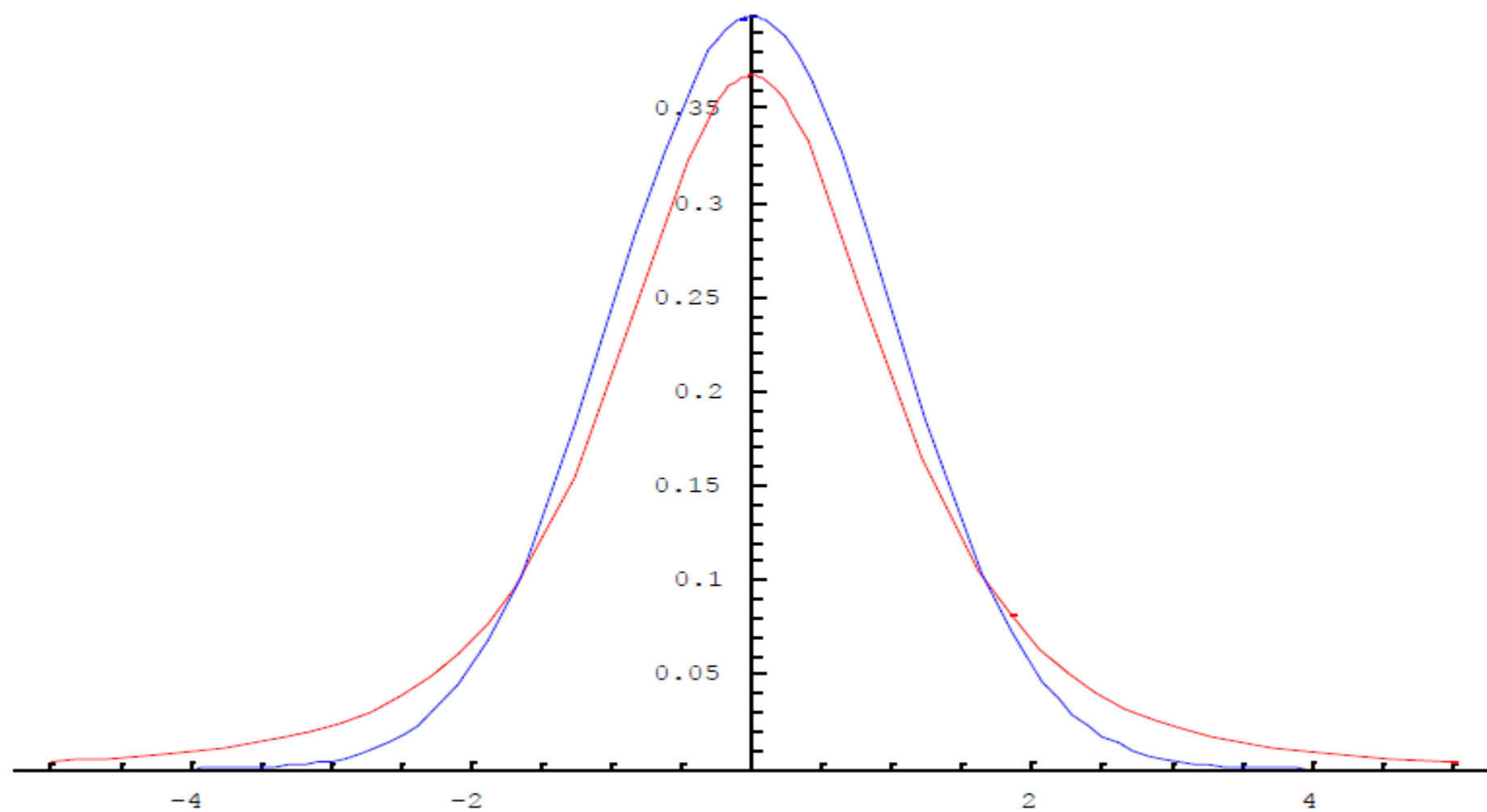
Red: t distribution Blue: standard normal curve

Student T Distribution with 2 degrees of freedom



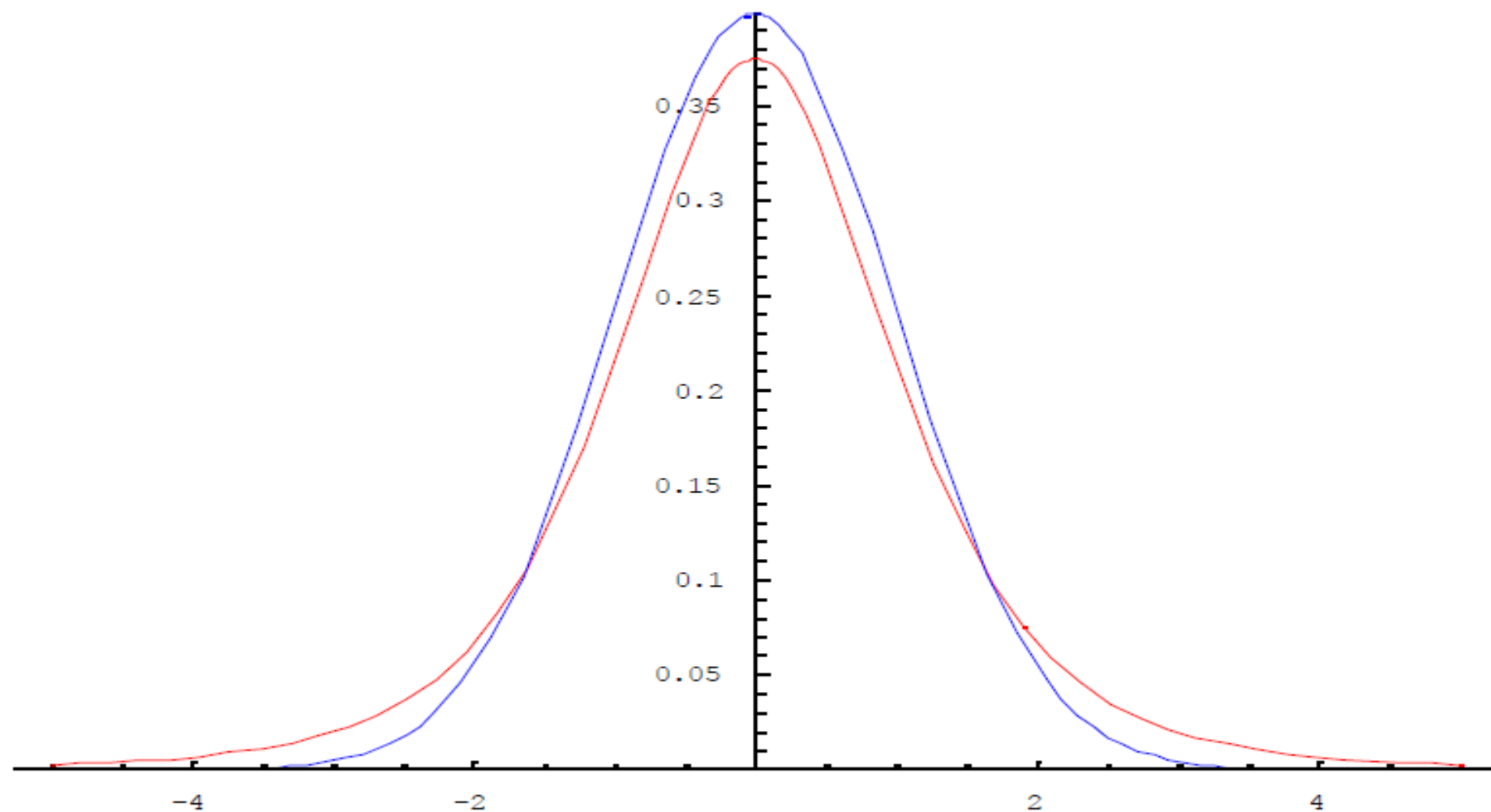
Red: t distribution Blue: standard normal curve

Student T Distribution with 3 degrees of freedom



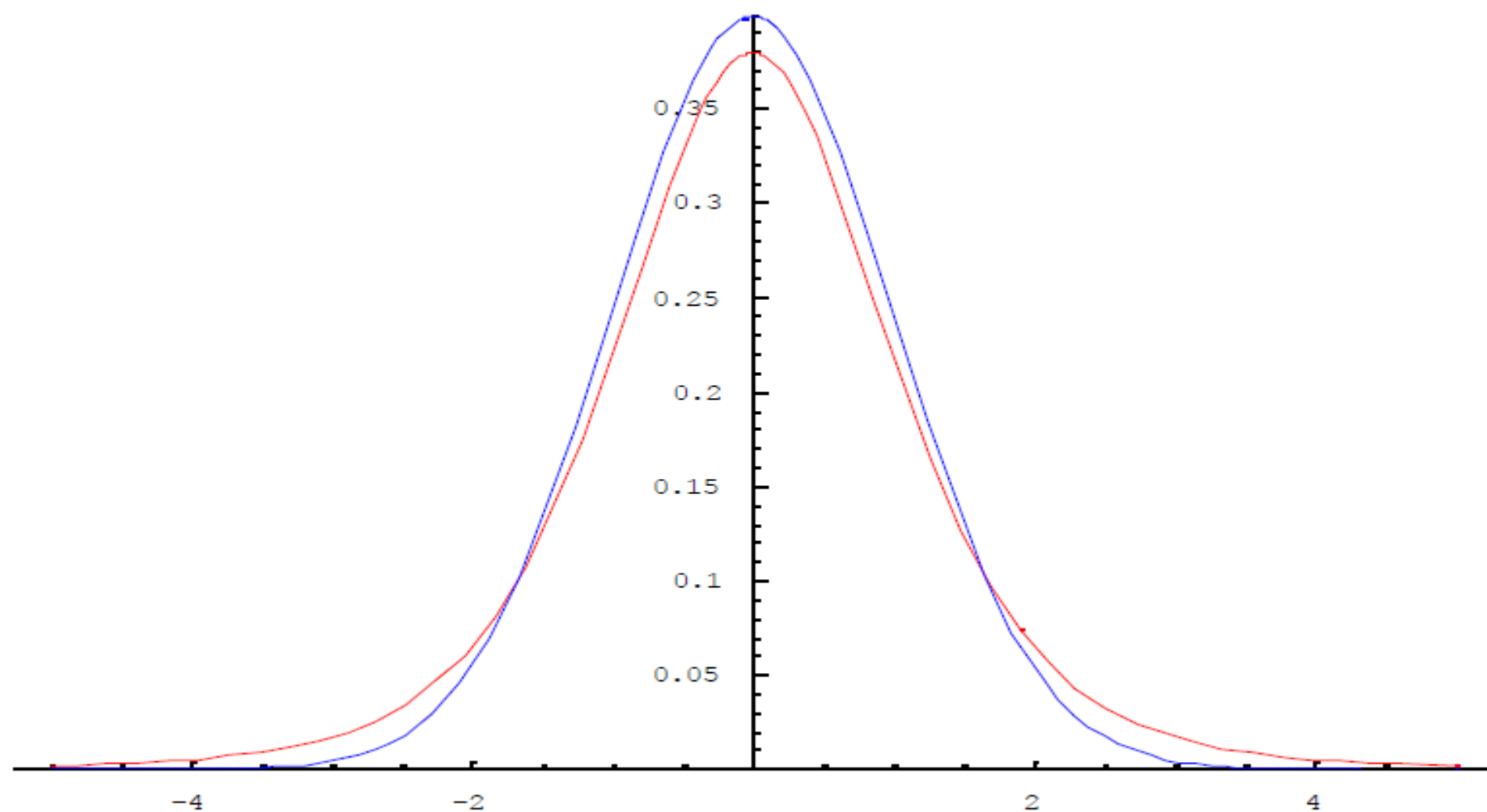
Red: t distribution Blue: standard normal curve

Student T Distribution with 4 degrees of freedom



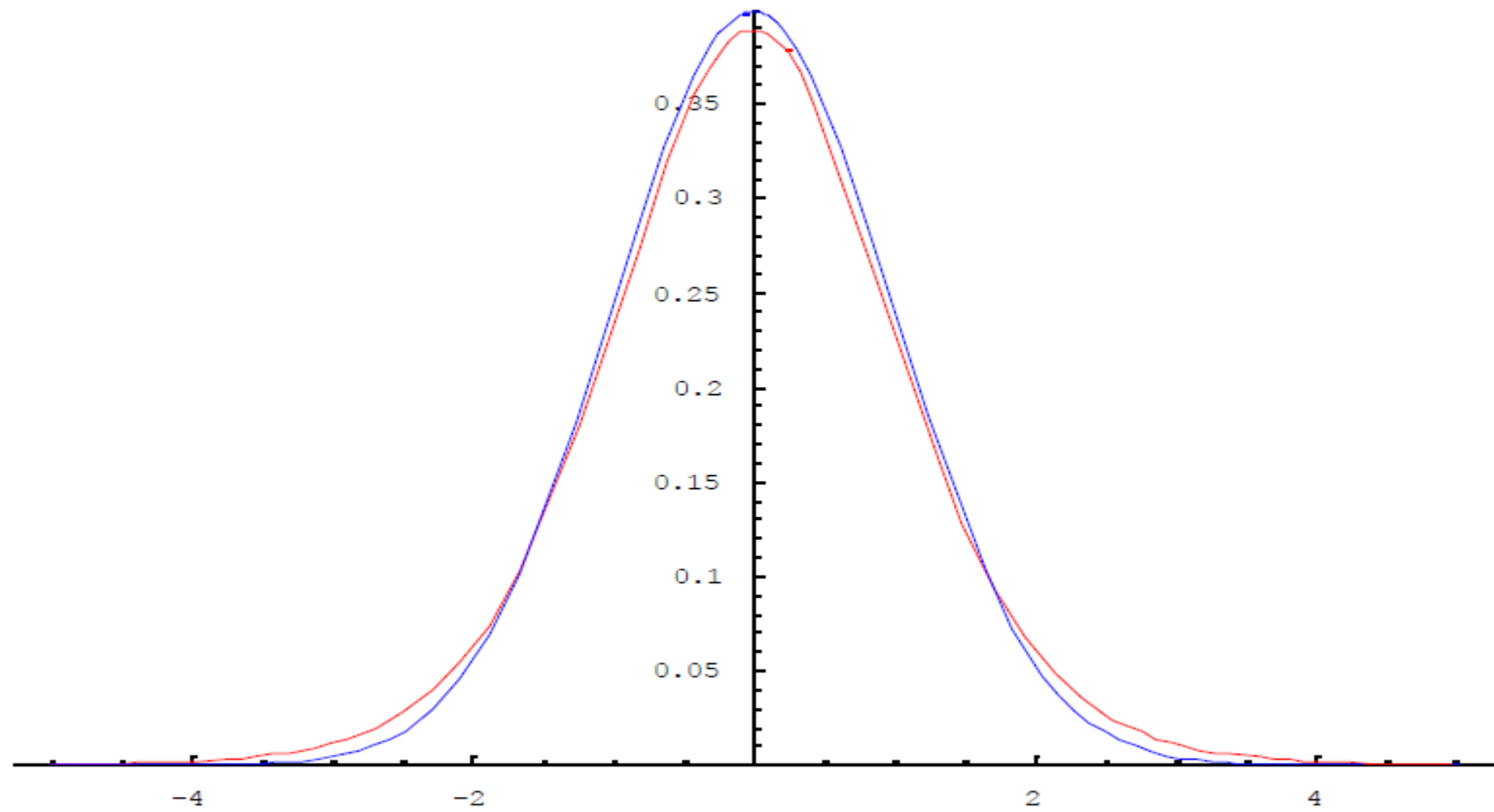
Red: t distribution Blue: standard normal curve

Student T Distribution with 5 degrees of freedom



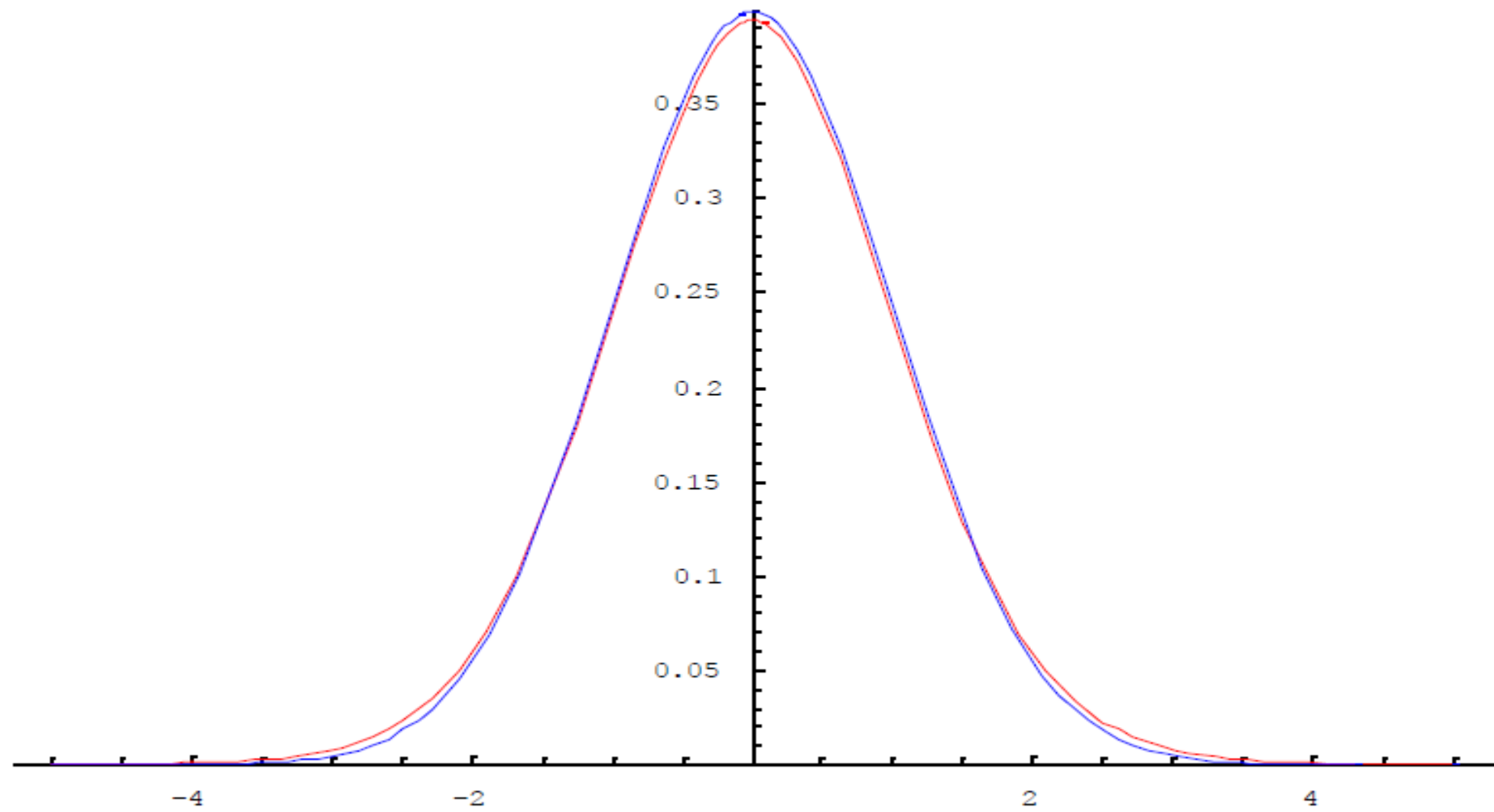
Red: t distribution Blue: standard normal curve

Student T Distribution with 10 degrees of freedom



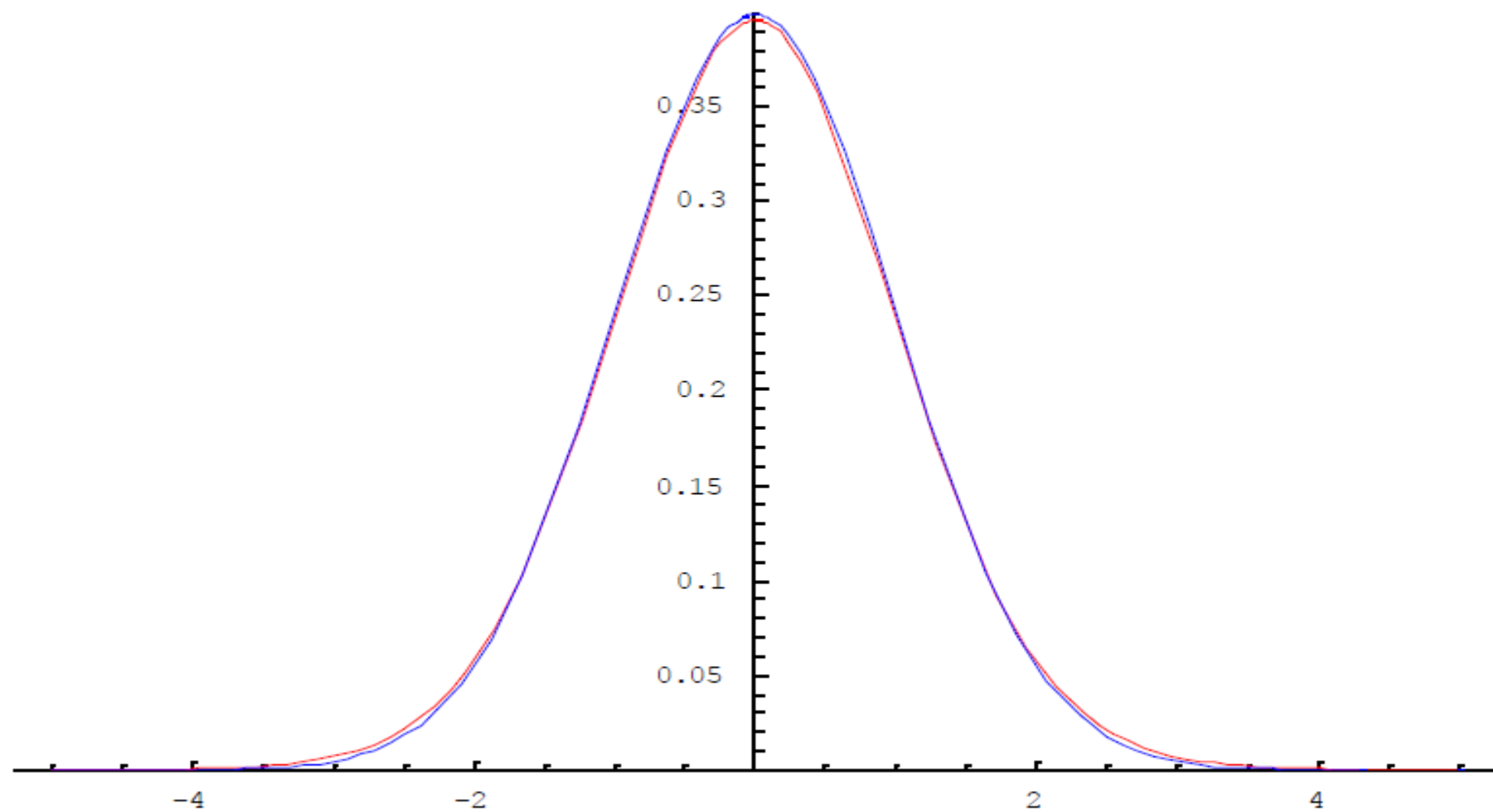
Red: t distribution Blue: standard normal curve

Student T Distribution with 20 degrees of freedom



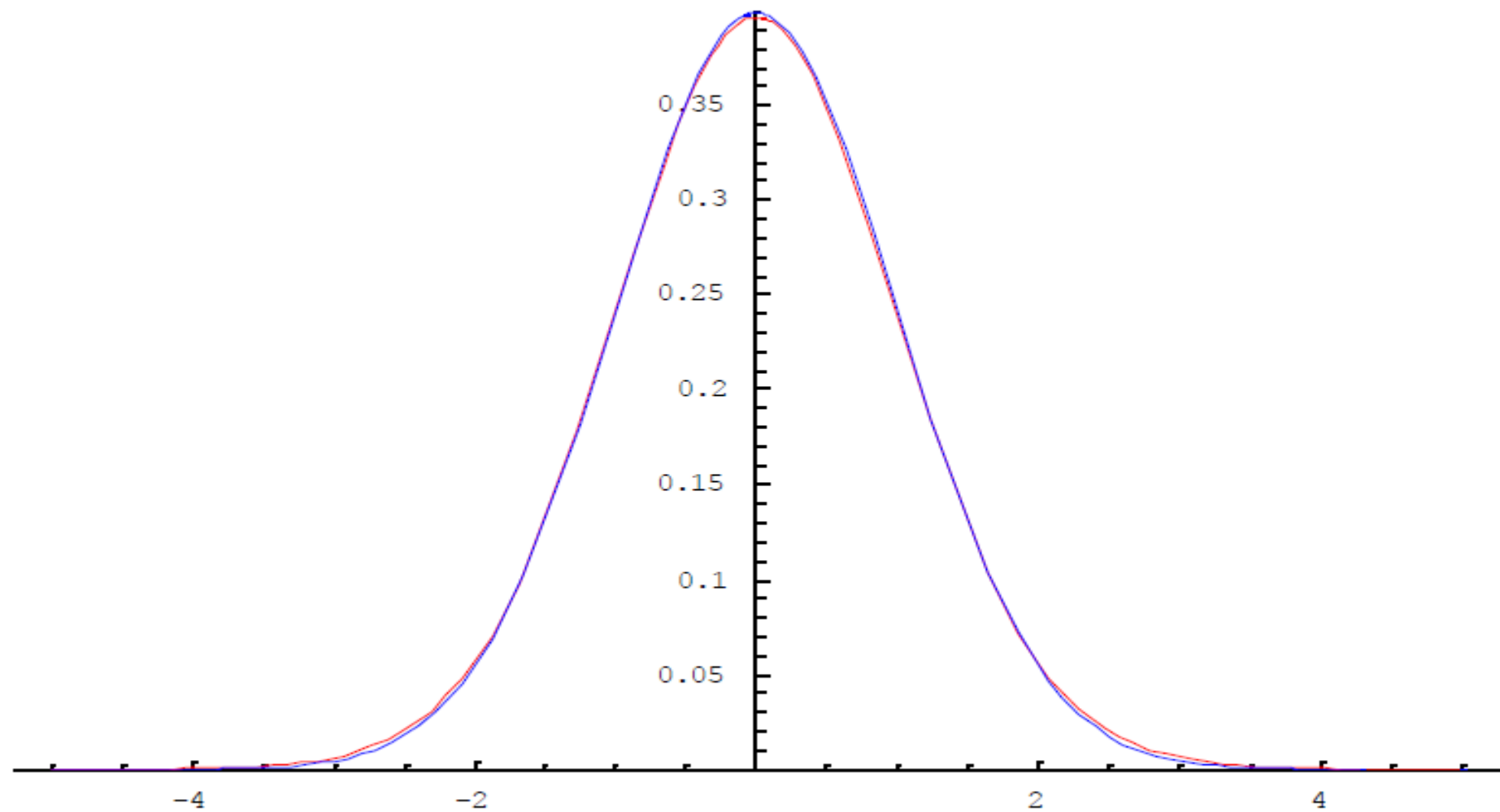
Red: t distribution Blue: standard normal curve

Student T Distribution with 30 degrees of freedom



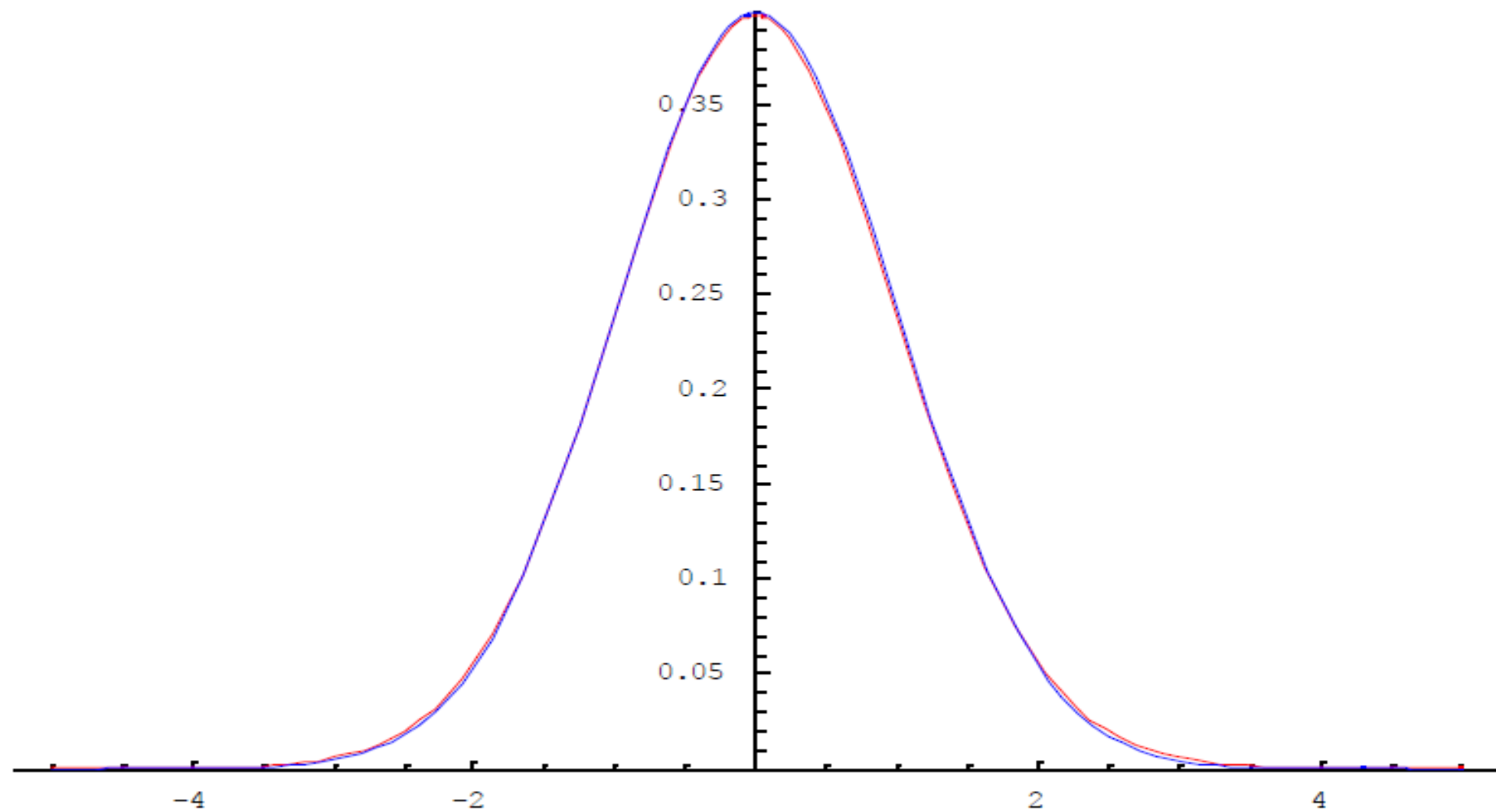
Red: t distribution Blue: standard normal curve

Student T Distribution with 40 degrees of freedom



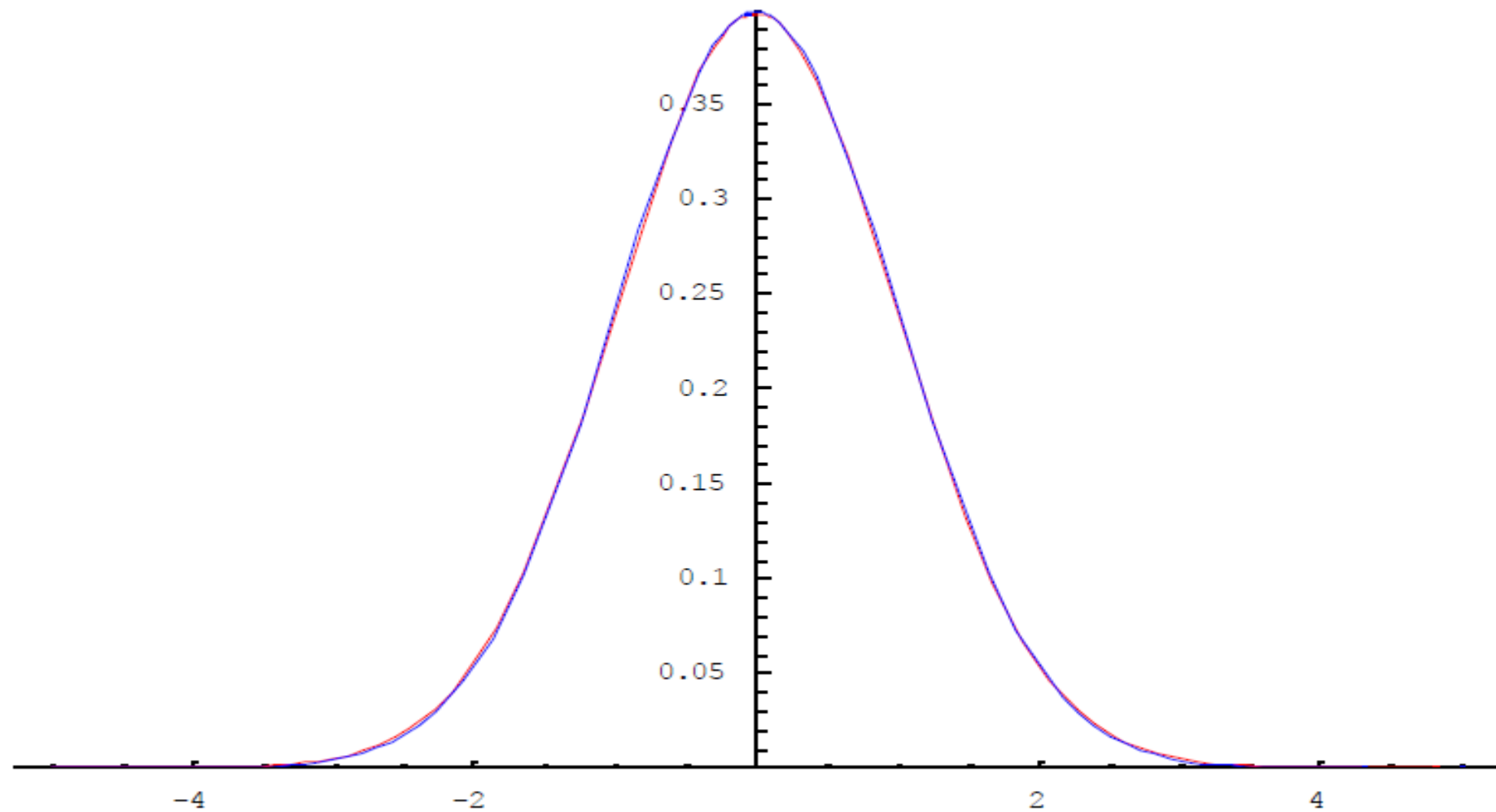
Red: t distribution Blue: standard normal curve

Student T Distribution with 50 degrees of freedom



Red: t distribution Blue: standard normal curve

Student T Distribution with 100 degrees of freedom



Red: t distribution Blue: standard normal curve

Applications of t-test:

- 1) For testing the significance of the difference between sample mean and population mean.
- 2) For testing the difference between means of two samples.
- 3) For testing significance of the observed correlation co-efficient.
- 4) For testing the significance of observed regression co-efficient.

1. Testing a Claim about a Mean

1. Test of Significance of a Mean of a Small Sample:

- To test the hypothesis that there is no significant difference between sample mean and assumed mean μ_0 of the population.
- $H_0: \mu = \mu_0$ vs $H_1: \mu \neq \mu_0$
- Choose value of t with an appropriate degree of freedom $n-1$ and at a required LoS

$$t_{cal} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

- Calculate test statistic
- If $|t_{cal}| < |t_{tab}|$ the H_0 may be accepted.
- If $t_{0.05}$ is the tabulated value of t for $v = (n - 1)$ d.f. at 5% los, then the 95% confidence limits for μ are given by :

$$\left(\bar{x} - t_{0.05} \frac{s}{\sqrt{n}}, \quad \bar{x} + t_{0.05} \frac{s}{\sqrt{n}} \right)$$

Examples:

1. A random sample of 10 boys had the following I.Q.'s: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100. Do these data support the assumption of a population mean I.Q. of 100 ? Also, find a reasonable range in which most of the mean I.Q. values of samples of 10 boys lie.

Solution:

Population is a data collection of I.Q values of boys.

Given that, a random sample of 10 boys had the following I.Q.'s: 70, 120, 110, 101, 88, 83, 95, 98, 107, 100.

So, sample size is 10 and assumed mean I.Q of boys is 100, i.e., $n=10$ and $\mu_0=100$.

➤ Thus, the present problem about single mean of small samples. Hence, t-test can be used.

Test of significance of single mean:

1. State H_0

The data are consistent with the assumption of a mean I.Q. of 100 in the population, i.e., $H_0: \mu = 100$.

2. State H_1

$$H_1: \mu \neq 100.$$

3. Choose LoS $\alpha \rightarrow \alpha = 0.05$

4. Set Up Critical Value(s)

Tabulated value of statistic “t” at 5% LoS and with (10 - 1) i.e., 9 d.f for two-tailed test is 2.262. i.e. $t_{0.05} = 2.262$

5. Compute Test Statistic:

Under H_0 the test statistic for single mean is

$$t_{cal} = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

Where sample mean and S.D of sample are to be computed from the sample values of I.Q. ·s.

Test of significance of single mean:

Sample values	Deviation=Sample values-Sample mean	Square of deviation
70	-27.2	739.84
120	22.8	519.84
110	12.8	163.84
101	3.8	14.44
88	-9.2	84.64
83	-14.2	201.64
95	-2.2	4.84
98	0.8	0.64
107	9.8	96.04
100	2.8	7.84
Total 972		183.60

$$n = 10, \bar{x} = \frac{972}{10} = 97.2 \text{ and}$$

$$s^2 = \frac{183.60}{10-1} = 203.73$$

$$\therefore t = \frac{97.2 - 100}{\sqrt{203.73/10}} = \frac{-2.8}{\sqrt{20.37}}$$

$$= -0.62$$

$$|t_{cal}| = 0.62$$

6. Make Statistical Decision:

Since $|t_{cal}| < |t_{tab}|$. So, H_0 may be accepted at 5% level of significance.

we may conclude that the data are consistent with the assumption of mean I.Q. of 100 in the population.

Confidence limits for population mean:

- The 95% confidence limits within which the mean I.Q. values of samples of 10 boys will lie are given by

$$\begin{aligned} & \left(\bar{X} - t_{0.05} \frac{s}{\sqrt{n}}, \bar{X} + t_{0.05} \frac{s}{\sqrt{n}} \right) \\ &= (97.2 - 2.262 \times 4.514, 97.2 + 2.262 \times 4.514) \\ &= (86.99, 107.41) \end{aligned}$$

Hence the required 95% confidence interval is [86.99, 107.41].

2. Testing a claim about a difference of means

2. t-Test for Difference of Means:

- Suppose we want to test if two independent samples of sizes
- n_1 and n_2 have been drawn from two normal populations with means μ_x and μ_y , respectively.
- Under the null hypothesis $H_0: \mu_x = \mu_y$ and assumption that both the population have the same variance.
- Choose value of t with an appropriate degree of freedom $n_1 + n_2 - 2$ and at a required LoS.

- Calculate test statistic
- where

$$t_{cal} = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$\bar{x} = \frac{\sum x_i}{n_1}$$

$$\bar{y} = \frac{\sum y_j}{n_2}$$

$$S^2 = \frac{1}{n_1 + n_2 - 2} \left[\sum (x_i - \bar{x})^2 + \sum (y_j - \bar{y})^2 \right]$$

Examples:

- In a certain experiment to compare two types of pig foods A and B. The following results of increase in weights (in lbs) were observed in pigs. Assuming that the two samples of pigs are independent, *can we conclude that food B is better than food A?*

Food A	25	32	30	34	24	14	32	24	30	31	35	25			
Food B	44	34	22	10	47	31	40	30	32	35	18	21	35	29	22

Solution:

Here, we have two populations indicating weights of pigs due to food A and B respectively.

Given that, *the drawn samples are independent and assume that both population has same variance. Samples sizes are 12 and 15*

➤ *Thus, the present problem about difference of means of small samples. Hence, t-test can be used.*

Test of significance of difference of means:

1. State H_0

there is no significant difference in increase in weights due to diets A and B i.e.,
 $H_0 : \mu_x = \mu_y$.

2. State H_1

$H_1 : \mu_x < \mu_y$ (Left-tailed).

3. Choose LoS $\alpha \rightarrow \alpha = 0.05$

4. Set Up Critical Value(s)

Tabulated value of statistic “t” at 5% LoS and with (12+15-2) i.e., 25 d.f for left tailed test is **-1.708**. i.e $t_{0.05} = -1.708$

5. Compute Test Statistic:

Under H_0 the test statistic for difference of means is

$$t_{cal} = \frac{(\bar{x} - \bar{y}) - (\mu_x - \mu_y)}{S \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

Where sample mean and S.D of sample are to be computed from the sample values of weight's

Computations of test statistics:

Diet A			Diet B		
X	$X - \bar{X}$	$(X - \bar{X})^2$	Y	$Y - \bar{Y}$	$(Y - \bar{Y})^2$
25	-3	9	44	14	196
32	4	16	34	4	16
30	2	4	22	-8	64
34	6	36	10	-20	400
24	-4	16	47	17	289
14	-14	196	31	1	1
32	4	16	40	10	100
24	-4	16	30	0	0
30	2	4	32	2	4
31	3	9	35	5	25
35	7	49	18	-12	144
25	-3	9	21	-9	81
			35	5	25
			29	-1	1
			22	-8	64
336	0	380	450	0	1410

Here

$$\left. \begin{array}{l} n_1 = 12, \\ \Sigma x = 336 \\ \Sigma(x - \bar{x})^2 = 380 \end{array} \right\} \text{ and } \left. \begin{array}{l} n_2 = 15 \\ \Sigma y = 450 \\ \Sigma(y - \bar{y})^2 = 1410 \end{array} \right\}$$

$$\therefore S^2 = \frac{1}{n_1 + n_2 - 2} [\Sigma(x - \bar{x})^2 + \Sigma(y - \bar{y})^2] = 71.6$$

$$\therefore t = \frac{\bar{x} - \bar{y}}{\sqrt{S^2 \left(\frac{1}{n_1} + \frac{1}{n_2} \right)}} = \frac{28 - 30}{\sqrt{71.6 \left(\frac{1}{12} + \frac{1}{15} \right)}} = \frac{-2}{\sqrt{10.74}} = -0.609$$

6. Make Statistical Decision:

Since $|t_{\text{cal}}| < |t_{\text{tab}}|$. So, H_0 may be accepted at 5% level of significance.

3. Chi-square test for goodness of fit & independence of attributes.

3. A. Test of Goodness of Fit:

- A very powerful test for testing the significance of the discrepancy between theory and experiment.
- If O_i ($i = 1, 2, \dots, n$) is a set of observed frequencies and E_i ($i = 1, 2, \dots, n$) is the corresponding set of expected frequencies, then Karl Pearson's chi-square, given by

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right] \quad \sum_{i=1}^n O_i \approx \sum_{i=1}^n E_i = N$$

- Follows chi-square distribution with $(n-1)$ degree's freedom

Examples:

The following table gives the number of aircraft accidents that occurs during the various days of the week. Find whether the accidents are uniformly distributed over the week.

Days	Sun	Mon	Tues	Wed	Thus	Fri	Sat
No. of accidents	14	16	8	12	11	9	14

(Given: the .values of chi-square significant at 5, 6, 7, df. are respectively' 11.07, 12.59: 14.07 at the 5% level of significance.)

Solution:

1. State H_0

the accidents are uniformly distributed over the week.

Under the null hypothesis, the expected frequencies of the accidents on each of the days would be :

Days	Sun	Mon	Tues	Wed	Thus	Fri	Sat	Total
No. of accidents	12	12	12	12	12	12	12	84

2. State H_1

*the accidents are **not** uniformly distributed over the week.*

3. Choose LoS $\alpha \rightarrow \alpha = 0.05$

4. Set Up Critical Value(s)

Tabulated value of statistic “ χ^2 ” at 5% LoS and with (7-1) i.e., 6 d.f is **12.59**.

i.e $\chi^2_{0.05} = 12.59$

5. Compute Test Statistic:

Under H_0 the test statistic for goodness of fit is

$$\chi^2 = \sum_{i=1}^n \left[\frac{(O_i - E_i)^2}{E_i} \right] \quad \sum_{i=1}^n O_i \simeq \sum_{i=1}^n E_i = N$$

$$\begin{aligned}\chi^2 &= \frac{(14 - 12)^2}{12} + \frac{(16 - 12)^2}{12} + \frac{(8 - 12)^2}{12} + \frac{(12 - 12)^2}{12} \\ &\quad + \frac{(11 - 12)^2}{12} + \frac{(9 - 12)^2}{12} + \frac{(14 - 12)^2}{12} \\ &= \frac{1}{12} (4 + 16 + 16 + 0 + 1 + 9 + 4) = \frac{50}{12} \\ &= 4.17\end{aligned}$$

6. Make Statistical Decision:

Since the calculated χ^2 is much less than the tabulated value, it is highly insignificant and we accept the null hypothesis.

Hence we conclude that the accidents are uniformly distributed over the week.

3. B. Chi-Square Test for Independence of Attributes:

- This test is applied when you have two categorical variables from a single population.
- It is used to determine whether there is a significant association between the two variables.

For example, A tea company appoints 4 Sales men A,B,C,D and observes their sales in three (3) seasons summer, winter, monsoon, the figures given in lakhs.

Seasons	Sales man				Seasons Total
Summer	A 36	B 36	C 21	D 35	128
Winter	28	29	31	32	120
Monsoon	26	28	29	29	112
Salesman's Total	90	93	81	96	360

3. B Conts:

- State the Hypotheses:

Suppose that Variable A has r levels, and Variable B has c levels.

- **Ho: Variable A and Variable B are independent.**
- H_1 : Variable A and Variable B are not independent.
- **Expected frequencies.** Compute $r \times c$ expected frequencies, according to the following formula $E_{r,c} = (n_r \times n_c) / N$
- **Test statistic.**
- The test statistic is a chi-square “ χ^2 ” variable defined by the following equation.
$$\chi^2 = \sum \left[\frac{(O_{r,c} - E_{r,c})^2}{E_{r,c}} \right] \quad \sum O_{r,c} = \sum E_{r,c} = N$$
- Follows chi-square distribution with $(r - 1) \times (c - 1)$ degree's freedom

Examples:

From the following table, showing the number of plants having certain characters, test the hypothesis that the flower color is independent of flatness of leaf.

	Flat leaves	Curled leaves	Total
White Flowers	99	36	135
Red Flowers	20	5	25
Total	119	41	160

Solution:

1. State H_0

The flower color is independent of flatness of leaf.

2. State H_1

The flower color is dependent of flatness of leaf.

3. Choose LoS $\alpha \rightarrow \alpha = 0.1$

4. Set Up Critical Value(s)

Tabulated value of statistic “ χ^2 ” at 10% LoS and with $d.f=(2-1) \times (2-1)$, i.e., 1 d.f is **0.0158**

i.e $\chi^2_{0.1} = 0.0158$ under 1 degree of freedom

5. Compute Test Statistic:

the following table shows the theoretical frequencies.

	0	1	
	<i>Flat leaves</i>	<i>Curled leaves</i>	<i>Total</i>
White flowers	$\frac{135 \times 119}{160} = 100$	$\frac{135 \times 41}{160} = 35$	135
Red flowers	$\frac{25 \times 119}{160} = 19$	$\frac{25 \times 41}{160} = 6$	25
Total	119	41	160

Computations of test statistics:

Under the null hypothesis, the test statistic is

Observed frequencies	Expected frequencies
99	100
36	35
20	19
5	6

$$\chi^2 = \sum \left[\frac{(O_{r,c} - E_{r,c})^2}{E_{r,c}} \right] \quad \sum O_{r,c} = \sum E_{r,c} = N$$
$$= \frac{(99 - 100)^2}{100} + \frac{(36 - 35)^2}{35} + \frac{(20 - 19)^2}{19} + \frac{(5 - 6)^2}{6}$$
$$= \frac{1}{100} + \frac{1}{35} + \frac{1}{19} + \frac{1}{6} = 0.2579$$

6. Make Statistical Decision:

Since the calculated χ^2 is greater than the tabulated value, it is highly significant and we reject the null hypothesis.

Hence we conclude that the flower color is dependent of flatness of leaf at the 0.1 level of significance.

F-Test for Equality of Population Variances

- Suppose we want to test
 - (i) Whether two independent samples $x_1, x_2, \dots, x_{n_1}$ and $y_1, y_2, \dots, y_{n_2}$ have been drawn from the normal population with the same variance σ^2 .
 - (ii) Whether the two independent estimates of the population variance are homogeneous or not..
- Under the null hypothesis (H_0)
 - (i) Population variances are equal.
 - (ii) Two independent estimates of population variance are homogeneous, test *statistic F is given by*

$$F = \frac{S_x^2}{S_y^2}, S_x^2 > S_y^2$$

follows F-distribution with (n_1-1, n_2-1) degree's freedom

Examples:

Two random samples are as are :

<i>Sample</i>	<i>Size</i>	<i>Sum of squares of deviations from the mean</i>
<i>1</i>	<i>10</i>	<i>90</i>
<i>2</i>	<i>12</i>	<i>108</i>

- Test whether the samples come from the same normal population at 5% level of significance. [Given: $F_{0.05}(9, 11) = 2.90$, $F_{0.05}(11, 9) = 3.10$]
- Solution:

1. State H_0

The two samples have been drawn from the same normal population.

2. State H_1

The two samples not drawn from the same normal population.

3. **Given $\alpha \rightarrow \alpha = 0.05$**

4. **Set Up Critical Value(s)**

Tabulated value of statistic “ F ” at 5% LoS and with $d.f=(10-1, 12-1)$, i.e., (9, 11) d.f is **2.90**. Also, with $d.f=(12-1, 10-1)$, i.e., (11, 9) d.f is **3.10**

5. **Compute Test Statistic: Given**

$$n_1 = 10, \Sigma (x - \bar{x})^2 = 90$$

$$n_2 = 12, \Sigma (y - \bar{y})^2 = 108$$

$$S_1^2 = \frac{1}{n_1 - 1} \Sigma (x - \bar{x})^2 = \frac{90}{9} = 10$$

$$S_2^2 = \frac{1}{n_2 - 1} \Sigma (y - \bar{y})^2 = \frac{108}{11} = 9.82$$

$$F = \frac{S_1^2}{S_2^2} = \frac{10}{9.82} = 1.018$$

Tabulated $F_{0.05}(9, 11) = 2.90$

6. **Make Statistical Decision:**

Since the calculated F is less than tabulated F , it is not significant. Hence, null hypothesis of equality of population variances may be accepted.

Examples:

1. The mean weekly sales of soap bars in departmental stores was 146.3 bars per store. After an advertising campaign the mean weekly sales in 22 stores for a typical week increased to 153.7 and showed a standard deviation of 17.2. Was the advertising campaign successful?
2. *Test whether the color of the son's eyes is associated with that of the fathers by following contingency table,*

		<i>Eye colour in sons</i>		
		<i>not light</i>	<i>light</i>	
<i>Eye colour in fathers</i>	<i>Not light</i>	230	148	378
	<i>light</i>	251	471	622
		381	619	1000

Given: value of chi-square is 3.84 for 1 degree of freedom.